

Active Demo Selection for VLA Fine-Tuning

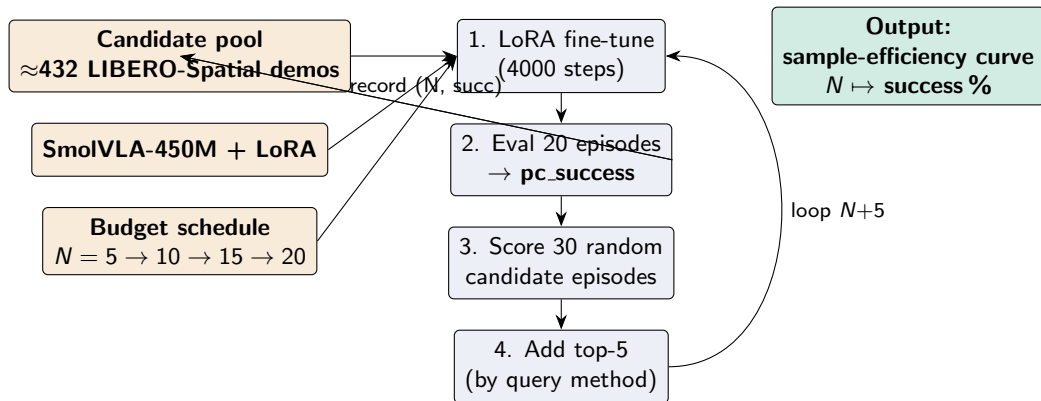
Project tracking deck — LIBERO/SmolVLA pipeline

anh0001 | branch research/conformal-active-demo-budgeting

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Question: which 20 demos should I label so a fine-tuned SmolVLA reaches the highest LIBERO-Spatial success rate?

Pipeline at a glance



4 rounds per cell. One cell = (method, suite, seed). ≈5 h wall on one RTX 6000.

Query method 1: random

- ▶ **Input signal:** none.
- ▶ **Scoring rule:** `score = rng.random()` per candidate.
- ▶ **Selection rule:** top-5 by score (i.e. uniform pick of 5 from 30).
- ▶ **Expected failure mode:** bad demos by chance; high seed variance.
- ▶ **Why it matters:** the de-facto baseline; if a learner doesn't beat this on a well-pretrained VLA, the acquisition signal is anti-informative.

Query method 2: conformal [BROKEN]

- ▶ **Input signal:** teacher-forced policy loss per candidate episode (12 stride-8 frames, 1 forward pass each).
- ▶ **Scoring rule (as implemented):** per candidate,
 1. add its own loss to the calibration buffer,
 2. return $\text{loss}/\hat{q}$ if $\text{buffer} \geq 20$, else raw loss.
- ▶ **Selection rule:** top-5 by score.
- ▶ **Failure mode (confirmed):** self-referential calibration + per-candidate threshold + sorted candidate order $\Rightarrow$ structurally picks *highest-indexed* episodes in the random subsample, not the most uncertain.
- ▶ See “Conformal bug” slide for the full diagnosis.

Query method 3: entropy

- ▶ **Input signal:** same teacher-forced loss as conformal.
- ▶ **Scoring rule:** score = mean loss over sampled frames.
- ▶ **Selection rule:** top-5 by score.
- ▶ **Expected failure mode:** loss is dominated by nuisance (trajectory length, contact noise, gripper switches), not epistemic uncertainty. Picks redundant “hard” demos.
- ▶ **Empirical:** trails random by 6-12 pp at $N=20$ (1 seed). Consistent with the nuisance hypothesis.

Query method 4: dispersion

- ▶ **Input signal:** $K=4$ samples from SmolVLA's flow-matching action head per frame; mean per-(step, dim) std across samples.
- ▶ **Scoring rule:** score = mean dispersion over sampled frames.
- ▶ **Selection rule:** top-5 by score.
- ▶ **Why this should help:** measures self-disagreement in *action space*, not next-token likelihood. On a saved checkpoint: dispersion stayed in $[0.037, 0.043]$ while teacher-forced loss spanned $[0.04, 8.69]$ across 4 episodes — loss is mostly nuisance.
- ▶ **Failure mode (also confirmed empirically):** regresses at $N=15$ ($29 \rightarrow 22\%$). Without coverage, dispersion overselects from one visually-busy task.

Query method 5: `dispersion_quota`

- ▶ **Input signal:** same as `dispersion`.
- ▶ **Scoring rule:** same as `dispersion`.
- ▶ **Selection rule:** greedy round-robin under per-task coverage. For each pick, choose the candidate from the task with the smallest (budget + already-picked) count; break ties by score.
- ▶ **Why this should help:** forces task diversity; prevents the redundant-task collapse that pure `dispersion` shows.
- ▶ **Status (1 seed, 3 rounds):** holds $29 \rightarrow 31\%$ at $N=15$ where pure `dispersion` drops to 22% ; within ~ 2 pp of random at $N=15$.

Current results: libero_spatial, max_demos = 20

N	rand s0	rand s1	conf s0	conf s1	ent s0	disp s0	disp+q s0
5	9.0	10.5	9.0	10.5	9.0	9.0	9.0
10	22.5	31.0	21.0	16.0	15.5	29.0	29.0
15	33.0	32.5	24.5	18.5	25.5	22.0	31.0
20	35.0	40.5	28.0	25.0	29.0	OOM	OOM

- ▶ Eval is *paired* across methods (env seed = 1000+ run seed). Gaps are not eval noise.
- ▶ Grayed conformal columns: implementation is broken; kept for reproducibility, not direction-claim.
- ▶ **OOM**: GPU co-tenant returned mid- $N=20$ -train; partial curve only.

Conformal bug — how scoring went wrong

Codex's invariant check:

if conformal score = $\text{loss}/\hat{q}$ with a *global* $\hat{q}$, ranking = entropy ranking.

Empirical: seed-0 $N=5 \rightarrow 10$ new picks

- ▶ conformal: {1394, 1447, 1458, 1493, 1523}
- ▶ entropy: {1394, 1458, 1567, 1574, 1579}

Overlap 2/5. **Invariant violated.**

Smoking gun in `active_loop.py:77{82:`

- ▶ Each candidate's own loss is added to the buffer *before* it is scored (self-reference).
- ▶ Buffer resets every round; `calib_min=20`, `max_candidates=30`.
- ▶ Candidates 0–19 return raw loss (~ 0.03); 20–29 return $\text{loss}/\hat{q}$ (~ 1.0). $20 \times$ scale gap.
- ▶ Top-5 always lands in the last ~ 10 iterated candidates = highest-indexed in the sorted subsample.

Implication: conformal in this top- k pool setting reduces to a monotone transform; ranking $\equiv$ entropy. The branding adds no acquisition signal.

Status & next decision

Done

- ▶ Phase S.0–S.5 brought up. PS.4 single-task gate passed (80 % on libero_spatial task 0).
- ▶ 5 methods $\times$ 1–2 seeds each, full curves for 5 cells; 2 cells partial ($N = 5/10/15$) due to OOM.
- ▶ Conformal bug found and committed in the tracker (05549edf).

Blocked

- ▶ Both GPUs occupied by an unrelated user (HMDB51 ConvGRU, RTX 6000 $\times 2$).
- ▶ Pending: dispersion/dispersion_quota re-runs to fill the $N=20$ point, then seed-1 of each method.

Next decision after the ablation completes

- ▶ If dispersion_quota cleanly beats random at $N=15-20$ across seeds: pivot the paper to “diversity-aware action-space uncertainty,” bury conformal as a negative-result diagnostic.
- ▶ If quota only matches random: the publishable story is the *acquisition-signal pathology* itself (loss is anti-informative for low-demo VLA), plus the corrected recipe.